

Pharmacotherapy 2: Ultimate Question Bank

Batch 5: Lecture 5 (Antiepileptic Drugs - AEDs)

50 High-Yield Exam Questions with Clinical Explanations

151. Which of the following correctly differentiates Simple Partial seizures from Complex Partial seizures?

- A) Simple partial seizures involve both hemispheres from the start
- B) Consciousness is preserved in simple partial, but impaired in complex partial seizures**
- C) Complex partial seizures only occur in children under 5
- D) Simple partial seizures always generalize into tonic-clonic seizures



Hint: As the doctor explained, in simple partial seizures, the patient does NOT lose consciousness. In complex partial, there is sensory hallucination, mental distortion, and loss of consciousness.

152. The most dramatic form of epilepsy, characterized by a loss of consciousness followed by stiffening and then jerking phases, is called:

- A) Absence (Petit mal) seizures
- B) Generalized Tonic-Clonic (Grand mal) seizures**
- C) Myoclonic seizures
- D) Febrile seizures



Hint: Tonic-clonic (Grand mal) involves a 'Tonic' (stiffening) phase followed by a 'Clonic' (jerking) phase, ending in exhaustion and confusion.

153. A young child frequently experiences brief, abrupt, and self-limiting staring spells with rapid eye-blinking lasting 3-5 seconds. This is characteristic of:

- A) Complex partial seizures
- B) Absence (Petit mal) seizures**
- C) Status Epilepticus
- D) Tonic-clonic seizures



Hint: The doctor noted that Absence seizures look like the patient is just 'staring' or 'daydreaming' briefly. It is highly dangerous depending on the patient's physical location (e.g., crossing a street).

154. Febrile seizures typically occur in children between 3 months and 5 years of age. How are they usually treated?

- A) Lifelong Phenytoin therapy
- B) High-dose Carbamazepine
- C) They are benign, self-limiting, and rarely require daily antiepileptic medication**
- D) Surgery

 **Hint:** The doctor emphasized that febrile seizures are provoked by high fever in young children. They are benign, resolve on their own as the child grows, and do not classify as chronic epilepsy requiring meds.

155. Which of the following drugs exerts its antiepileptic effect primarily by blocking voltage-gated Sodium (Na^+) channels?

- A) Ethosuximide
- B) Carbamazepine**
- C) Diazepam
- D) Tiagabine

 **Hint:** Carbamazepine, Phenytoin, and Lamotrigine all work by blocking Na^+ channels, inhibiting the generation and spread of repetitive action potentials.

156. Ethosuximide is uniquely effective for Absence seizures because its primary mechanism of action is:

- A) Enhancing GABA activity
- B) Blocking T-type Calcium (Ca^{++}) channels in the thalamus**
- C) Blocking Sodium (Na^+) channels
- D) Inhibiting Glutamate release

 **Hint:** Absence seizures are driven by T-type calcium currents. Ethosuximide (and Valproic Acid) specifically block these channels.

157. Benzodiazepines and Barbiturates control seizures primarily by:

A) Decreasing Glutamate release

B) Increasing the inhibitory effect of GABA on Chloride (Cl-) channels

C) Blocking Sodium channels

D) Blocking Calcium channels



Hint: They bind to the GABA-A receptor complex, increasing chloride influx, which hyperpolarizes the neuron and prevents firing.

158. What is the 'Golden Rule' regarding the choice and initiation of Antiepileptic Drugs (AEDs)?

A) Always start with a combination of at least 3 drugs

B) Start with polytherapy, then reduce to monotherapy

C) Treatment should always be initiated with Monotherapy (a single drug) first

D) Change the drug daily until seizures stop



Hint: The doctor stressed that Monotherapy is preferred to minimize compounding side effects (like severe sedation). A second drug is only added if the first completely fails.

159. What is the most dangerous consequence of abruptly stopping Antiepileptic Drug therapy, especially in a patient with tonic-clonic seizures?

A) Permanent memory loss

B) Precipitation of Status Epilepticus

C) Immediate hepatotoxicity

D) Severe osteoporosis



Hint: The doctor literally yelled 'DON'T STOP!' in the lecture. Abrupt withdrawal removes the seizure threshold protection, causing continuous, life-threatening seizures (Status Epilepticus).

160. A major adverse effect shared by almost all AEDs, which can negatively impact a child's academic performance, is:

A) Renal failure

B) Sedation and Drowsiness

C) Hair loss

D) Gum hypertrophy



Hint: The doctor highlighted that CNS depression (sedation, drowsiness, poor concentration) is a major issue, often making epileptic children struggle academically.

161. Carbamazepine is considered the drug of first choice for which types of seizures?

A) Absence seizures only

B) All partial seizures and Generalized tonic-clonic seizures

C) Myoclonic seizures

D) Febrile seizures



Hint: Carbamazepine is highly effective for partial (simple/complex) and tonic-clonic seizures by blocking Na⁺ channels.

162. Which type of seizure can actually be worsened or exacerbated by the administration of Carbamazepine?

A) Simple partial seizures

B) Complex partial seizures

C) Absence seizures

D) Tonic-clonic seizures



Hint: Carbamazepine is strictly contraindicated in Absence seizures. You must use Ethosuximide or Valproic Acid instead.

163. A unique pharmacokinetic property of Carbamazepine is that it:

- A) Is excreted 100% unchanged in the urine
- B) Induces its own drug-metabolizing enzymes in the liver (Auto-induction)**
- C) Inhibits liver enzymes, raising the levels of other drugs
- D) Is not absorbed orally



Hint: As an 'Enzyme Inducer', it speeds up the liver's CYP450 system. It induces the metabolism of itself and other drugs (like oral contraceptives), decreasing their effectiveness.

164. Chronic administration of Carbamazepine produces an epoxide metabolite that is highly associated with which severe toxicity?

- A) Gingival hyperplasia
- B) Leukopenia and Aplastic Anemia**
- C) Nephrotoxicity
- D) Pancreatitis



Hint: The doctor stressed that patients on Carbamazepine require frequent blood tests to monitor for fatal bone marrow suppression (aplastic anemia) and liver toxicity.

165. Which of the following drugs acts as an Enzyme INHIBITOR, significantly raising the plasma levels of other co-administered AEDs?

- A) Carbamazepine
- B) Phenobarbital
- C) Valproic Acid (Sodium Valproate)**
- D) Phenytoin



Hint: While Carbamazepine and Phenytoin are INDUCERS, Valproic Acid is a potent INHIBITOR of hepatic microsomal enzymes.

166. Valproic Acid is highly bound to plasma proteins (90%). What happens if a patient taking Valproic Acid also takes Aspirin?

- A) Nothing happens
- B) Aspirin displaces Valproic Acid, leading to Valproate toxicity**
- C) Valproic Acid prevents Aspirin absorption
- D) Aspirin induces the metabolism of Valproic Acid

 **Hint:** Drugs that are highly protein-bound compete for Albumin. Aspirin can displace Valproic Acid (or Phenytoin), increasing the 'free' active drug in the blood and causing sudden toxicity.

167. Valproic acid is considered a 'Broad-Spectrum' antiepileptic because its mechanism involves:

- A) Only blocking Na⁺ channels
- B) Only blocking Ca⁺⁺ channels
- C) Blocking Na⁺ channels, blocking T-type Ca⁺⁺ channels, and increasing GABA levels**
- D) Decreasing Glutamate release only

 **Hint:** Because it attacks epilepsy from three different angles, it is effective against almost all types of seizures (Partial, Tonic-Clonic, and Absence).

168. Which of the following is a severe, life-threatening adverse effect of Valproic Acid that requires frequent monitoring?

- A) Renal failure
- B) Hepatic toxicity (Hepatotoxicity)**
- C) Pulmonary fibrosis
- D) Cardiac arrest

 **Hint:** Valproic acid can cause fatal hepatotoxicity, especially in young children on multiple drugs. Liver enzyme monitoring is mandatory.

169. A pregnant woman taking Valproic Acid is at a high risk of having a baby with:

A) Cleft palate

B) Spina Bifida (Neural tube defects)

C) Phocomelia

D) Down syndrome



Hint: Valproate is notoriously teratogenic. It causes Spina Bifida (a failure of the neural tube to close). It should be heavily avoided in pregnancy if possible.

170. A common, yet harmless, cosmetic side effect of Valproic Acid is:

A) Blue skin discoloration

B) Transient hair loss (Alopecia)

C) Excessive body hair growth

D) Dental caries



Hint: Valproate is known to cause alopecia (hair loss) and weight gain, which can be distressing to patients.

171. Phenytoin acts by stabilizing excitable membranes (inhibiting Na⁺ influx). It is highly unique because at normal doses it:

A) Causes profound coma

B) Is NOT a generalized CNS depressant

C) Acts as a strong muscle relaxant

D) Cures schizophrenia



Hint: Unlike barbiturates or benzodiazepines, Phenytoin suppresses seizures without causing generalized CNS depression (lethargy/hypnosis) at normal doses.

172. Because Phenytoin stabilizes all excitable membranes, it is also clinically used to treat:

- A) Asthma
- B) Peptic ulcers
- C) Arrhythmias, especially those induced by Digitalis toxicity**
- D) Hypertension

 **Hint:** Phenytoin stabilizes the heart's excitable membranes just like it stabilizes the brain's, making it an effective antiarrhythmic for digitalis toxicity.

173. A very common side effect of Phenytoin, especially in children, that requires good dental hygiene is:

- A) Tooth decay
- B) Gingival hyperplasia (overgrowth of gums)**
- C) Jaw bone loss
- D) Oral thrush

 **Hint:** Gingival hyperplasia is a hallmark side effect of Phenytoin. It can be minimized with excellent oral care.

174. Which of the following drugs exhibits 'Zero-Order Kinetics' at high doses, meaning a small dose increase can lead to sudden, severe toxicity?

- A) Gabapentin
- B) Levetiracetam
- C) Phenytoin**
- D) Ethosuximide

 **Hint:** Phenytoin's metabolizing enzymes saturate quickly. Once saturated, clearance becomes zero-order (constant amount, not percentage), causing plasma levels to skyrocket.

175. Topiramate reduces seizure activity via multiple mechanisms, including the inhibition of Carbonic Anhydrase. This specific mechanism increases the risk of:

- A) Liver failure
- B) Renal stones**
- C) Gastric ulcers
- D) Aplastic anemia

 **Hint:** Carbonic anhydrase inhibition alters urine pH and citrate excretion, promoting the formation of kidney stones.

176. A patient on oral contraceptives is prescribed Topiramate for epilepsy. What is a crucial counseling point?

- A) Topiramate will cure her epilepsy in a week
- B) She must stop eating carbohydrates
- C) Topiramate decreases the concentration of ethinyl estradiol, so she needs a higher dose or backup contraception**
- D) The combination causes severe hypertension

 **Hint:** Topiramate induces enzymes that clear estrogen faster, potentially leading to unintended pregnancy if the contraceptive dose isn't adjusted.

177. Topiramate can cause 'Oligohydrosis'. What does this mean clinically?

- A) Excessive sweating
- B) Decreased sweating, which can lead to dangerous Hyperthermia**
- C) Excessive urination
- D) Decreased tear production

 **Hint:** Oligohydrosis (decreased sweating) impairs the body's ability to cool itself, risking hyperthermia especially in children during hot weather.

178. In addition to treating epilepsy, Topiramate is officially approved and highly effective for:

A) Gout prophylaxis

B) Migraine headache prophylaxis

C) Peptic ulcer disease

D) Schizophrenia



Hint: Topiramate and Valproic acid are anticonvulsants that are also heavily used to prevent migraine attacks.

179. Zonisamide is an AED that blocks Na^+ and Ca^{++} channels. However, it is chemically a Sulfonamide derivative. What is the clinical implication?

A) It is an antibiotic

B) It is absolutely contraindicated in patients with severe Sulfa allergies

C) It cannot be given with water

D) It causes immediate liver failure



Hint: Patients with a known hypersensitivity to sulfa drugs (like sulfamethoxazole) may have a severe cross-reaction to Zonisamide.

180. Which of the following newer AEDs is remarkably free of pharmacokinetic drug interactions, making it an excellent choice for adjunctive (add-on) therapy?

A) Carbamazepine

B) Phenytoin

C) Levetiracetam

D) Valproic Acid



Hint: Levetiracetam (Keppra) is unique. It doesn't bind to plasma proteins and doesn't induce/inhibit CYP450, making it incredibly safe to add to other drugs.

181. Tiagabine is an adjunctive therapy for partial seizures. Its highly specific mechanism of action is:

A) Blocking NMDA receptors

B) Blocking GABA uptake into presynaptic neurons (GAT-1 inhibitor), thereby enhancing GABA activity

C) Opening Sodium channels

D) Inhibiting Carbonic Anhydrase



Hint: By preventing the reuptake of GABA, Tiagabine leaves more of this inhibitory neurotransmitter in the synapse to calm the brain.

182. Pregabalin and Gabapentin are structurally analogs of GABA, but their actual mechanism of action in epilepsy is:

A) Directly stimulating GABA receptors

B) Binding to voltage-gated Calcium channels to inhibit excitatory neurotransmitter release

C) Blocking Sodium channels

D) Inducing liver enzymes



Hint: Despite their names, they don't work on GABA receptors. They block calcium channels, reducing the release of glutamate and substance P.

183. Which of the following AEDs is primarily indicated for neuropathic pain associated with diabetic neuropathy in addition to partial seizures?

A) Ethosuximide

B) Pregabalin

C) Phenobarbital

D) Felbamate



Hint: Pregabalin (Lyrica) and Gabapentin are blockbuster drugs for nerve pain (neuropathy) caused by diabetes or shingles.

184. Which older AED is a barbiturate that induces its own metabolism (self-induction) and causes profound dependence?

A) Phenobarbital

- B) Phenytoin
- C) Carbamazepine
- D) Valproic acid



Hint: Phenobarbital is a powerful CNS depressant. It induces CYP450 enzymes (clearing itself and other drugs faster) and causes severe physical addiction.

185. Primidone is an AED that is partially metabolized into two active compounds. One of them is:

A) Phenytoin

B) Phenobarbital

- C) Carbamazepine
- D) Valproate



Hint: Primidone acts as a prodrug. The liver breaks it down into phenobarbital and phenylethylmalonamide, both of which have anticonvulsant activity.

186. Diazepam and Lorazepam (Benzodiazepines) are given intravenously (IV) as the drug of choice for the immediate treatment of:

A) Absence seizures

B) Status Epilepticus

- C) Migraines
- D) Peptic ulcers



Hint: Status Epilepticus is a medical emergency. IV Diazepam or Lorazepam acts immediately to boost GABA and stop the continuous seizure.

187. Felbamate blocks the NMDA receptor (reducing Glutamate excitation) but is strictly reserved for refractory epilepsies because it has a high risk of causing:

- A) Gingival hyperplasia
- B) Aplastic anemia and severe hepatic failure**
- C) Kidney stones
- D) Alopecia



Hint: Felbamate is a 'last resort' drug. Its use is heavily restricted because it can cause fatal bone marrow failure and liver necrosis.

188. Which of the following pairs of drugs are notorious Hepatic Enzyme INDUCERS?

- A) Valproic Acid and Ethosuximide
- B) Carbamazepine and Phenytoin**
- C) Levetiracetam and Gabapentin
- D) Topiramate and Tiagabine



Hint: Carbamazepine, Phenytoin, and Phenobarbital are the classic 'Inducers'. They speed up the liver, destroying other drugs (like oral contraceptives) too quickly.

189. A patient experiences blurred vision, nystagmus, diplopia, and ataxia (staggering gait). These cerebellar-vestibular manifestations are classic side effects of:

- A) Phenytoin or Carbamazepine toxicity**
- B) Valproic acid toxicity
- C) Levetiracetam
- D) Ethosuximide



Hint: Drugs that block Sodium channels in the brain (like Phenytoin and Carbamazepine) commonly cause these balance and visual disturbances when levels get too high.

190. Long-term use of Phenytoin and Carbamazepine can interfere with Vitamin D metabolism, leading to which adverse effect?

- A) Scurvy
- B) Osteoporosis**
- C) Anemia
- D) Blindness

 **Hint:** Enzyme inducers accelerate the breakdown of Vitamin D, reducing calcium absorption and causing bone thinning (osteoporosis) over time.

191. If Carbamazepine is administered alongside Erythromycin or Cimetidine, what is the likely outcome?

- A) Carbamazepine levels drop, causing seizures
- B) Erythromycin and Cimetidine INHIBIT Carbamazepine metabolism, leading to severe Carbamazepine toxicity**
- C) The drugs neutralize each other
- D) Erythromycin cures the epilepsy

 **Hint:** Erythromycin and Cimetidine are enzyme inhibitors. They block the liver from clearing Carbamazepine, causing it to build up to toxic levels.

192. Which drug requires the administration of Ethinyl Estradiol supplements for women using oral contraceptives, to avoid unwanted pregnancy?

- A) Gabapentin
- B) Topiramate**
- C) Valproic Acid
- D) Ethosuximide

 **Hint:** Topiramate (and other inducers like Carbamazepine) clears estrogen from the body too fast. The oral contraceptive pill will fail if the dose isn't increased.

193. A patient taking an AED complains of weight loss, numbness/tingling (paresthesia), and recently suffered a kidney stone. Which drug is he most likely taking?

- A) Valproic Acid
- B) Topiramate**
- C) Phenytoin
- D) Phenobarbital

 **Hint:** Weight loss and kidney stones (due to carbonic anhydrase inhibition) are the classic hallmark side effects of Topiramate.

194. Which of the following AEDs is heavily associated with causing WEIGHT GAIN as an adverse effect?

- A) Topiramate
- B) Zonisamide
- C) Valproic Acid**
- D) Ethosuximide

 **Hint:** Valproic Acid is famous for causing weight gain, alopecia, and hepatotoxicity. (Contrast with Topiramate which causes weight loss).

195. The NMDA receptor complex is activated by which EXCITATORY neurotransmitter, which some AEDs aim to block?

- A) GABA
- B) Dopamine
- C) Glutamic acid (Glutamate)**
- D) Serotonin

 **Hint:** Glutamate is the main excitatory neurotransmitter in the brain. Overactivity of Glutamate at the NMDA/AMPA receptors drives seizures.

196. Ethosuximide works ONLY on Absence seizures because:

A) It only blocks T-type calcium channels, which are exclusively responsible for the thalamocortical rhythms of absence seizures

- B) It is a massive GABA agonist
- C) It blocks sodium channels better than phenytoin
- D) It does not cross the blood-brain barrier



Hint: Absence seizures are generated by specific 'T-type' calcium currents in the thalamus. Ethosuximide surgically targets these channels and does nothing else.

197. Which of the following statements about Gabapentin is true?

- A) It is heavily metabolized by the liver
- B) It is tightly bound to plasma proteins
- C) It is not bound to plasma proteins and is excreted unchanged in the urine**
- D) It is an enzyme inducer



Hint: Gabapentin has zero drug-drug interactions because it ignores the liver completely and doesn't bind to proteins. It just goes straight out through the kidneys.

198. According to the Doctor, the ultimate goal of epilepsy treatment is:

- A) To use as many drugs as possible to guarantee success
- B) Complete control of seizures (seizure-free) preferably using Monotherapy to minimize toxicity**
- C) To keep the patient heavily sedated so they cannot seize
- D) To cure the genetic defect



Hint: The goal is a normal life. We want to stop the seizures entirely using just ONE drug if possible, avoiding the severe cognitive dulling and sedation of polytherapy.

199. If a patient has been seizure-free for 2-3 years, how should the physician handle the medication?

- A) Stop it abruptly on the 3rd anniversary
- B) Continue it for life without question
- C) Attempt a very gradual and slow withdrawal, monitoring closely for relapse**
- D) Double the dose to ensure it never comes back



Hint: The doctor explained that if a patient is stable for 2-3 years, the doctor may 'gradually' withdraw the drug over months to see if the patient can be drug-free.

200. Which of the following correctly pairs an AED with its classic teratogenic (fetal) malformation?

- A) Carbamazepine - Cataracts
- B) Valproic Acid - Spina Bifida (Neural tube defect)**
- C) Phenytoin - Phocomelia
- D) Topiramate - Blindness



Hint: Valproic acid interferes with folate metabolism in the fetus, preventing the spinal cord from closing properly (Spina Bifida).